Assignment no. 2

1) Prove that $\lim_{z \to z_0} \bar{z} = \bar{z}_0$ using $\varepsilon - \delta$ definition.

2) Find the values of

i) $(1+\sqrt{-3})^{3/4}$

ii) $(-i)^{1/6}$

iii) $(\sqrt{3} - i)^{24}$

3) Discuss the analyticity of the f^n $f(z) = z\bar{z} + \bar{z}^2$ in the complex plane.

(Hint $\frac{\partial f}{\partial \bar{z}} = 0$)

4) If $r^5 \cos k\theta + i r^k \sin k\theta$ is analytic, find the value of k.

5) Solve the following equations :-

i) $z^4 + 16 = 0$

ii) $z^2 = 1 - \sqrt{3}i$

6) If $|z| = 2$,

Prove that $|z^4 - 4z^2 + 3| \geq 3$

(Triangle Inequality)

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7) Discuss the continuity of the f^n

$$f(z) = \begin{cases} \dfrac{z^3 - 1}{z - 1}, & |z| \neq 1 \\ 3, & |z| = 1 \end{cases}$$

at the points $1, -1, i$ & $-i$.

8) For each of the following functions Determine the set of point at which it is i) differentiable ii) analytic. Find $f'(z)$ where it exists.

i) $\operatorname{sech} z$

iii) $f(z) = 6\bar{z}^2 - 2\bar{z} - 4i|z|^2$

ii) $f(z) = \dfrac{2z^2 + 6}{z(z^2+4)}$

9) Let $f: \mathbb{C} \to \mathbb{C}$ be defined by

$$f(z) = \begin{cases} \dfrac{\bar{z}^2}{z} & \text{if } z \neq 0 \\ 0 & \text{if } z = 0 \end{cases}$$

Show that C-R eq^ns are satisfied at $z = 0$

10) Suppose u is a harmonic f^n in a domain S. Show that if v is a harmonic conjugate of u, then v is also a harmonic conjugate of $u + c$, where c is any constant

Continued from
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Assignment I:
Topic - Complex Variables - I

Q. 1 Are the following functions analytic?

i) $z^2 + \frac{1}{z^2}$

ii) $f(z) = z|z|$

iii) $f(z) = \frac{x^2(1+i) - y^2(1-i)}{x+y}$, ~~$z \neq 0$~~ $x \neq -y$

$\quad\quad = 0$, ~~$z = 0$~~ otherwise

Q. 2 Show that following function is not analytic at the origin although Cauchy-Riemann equations are satisfied.

$f(z) = \frac{xy(y - ix)}{x^2 + y^2}$, $z \neq 0$

$\quad\quad 0$, $z = 0$

(Hint: Find $u_x(0,0)$, $u_y(0,0)$, $v_x(0,0)$, $v_y(0,0)$ explicitly).

Q. 3 Prove that if u is a real-valued harmonic function then $\frac{\partial u}{\partial z}$ is analytic.

Q. 4 Suppose that u is a real-valued harmonic function is domain D [$u: \mathbb{R}^2 \to \mathbb{R}$] & suppose that u^2 is also harmonic, prove that u is constant.

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Q.5 Find the analytic function whose real part is

$u = (x-1)^3 - 3xy^2 + 3y^2$

Q.6 Find the analytic function whose imaginary part is

$v = \dfrac{\sinh 2y}{\cos 2x + \cosh 2y}$

Q.7 Find the analytic function $f(z)$ whose real part is $r^2 \cos 2\theta - r \sin\theta$.

Q.8 Determine 'c' such that $u = \sin x . \cosh cy$ is harmonic & find a harmonic conjugate.

Q.9 An electrostatic field in the x-y plane is given by the potential function $\phi = x^2 - y^2 - 2xy - 2x + 3y$. Find the stream function. Find the complex potential function.

Q.10 Find the linear fractional transformation (LFT) which maps the points $0, i, -1$ of z-plane onto $i, 1, 0$ respectively of w-plane

Q.11 Find the image of the rectangle bounded by $x=0, y=0, x=2, y=3$ under the mapping $w = (1+i)z$

Q.12 Find all LFTs whose only fixed point is 0.

Q.13 Find all LFTs whose (only) fixed points are -2 & 2.

Q.14 Find the inverse $z = f^{-1}(w)$ for $w = f(z) = \dfrac{4z+i}{-3iz+1}$

Q.15 Find & sketch the image of the given region under the given mapping

i) $-\pi/4 < \operatorname{Arg} z < \pi/4$, $|z| < 1/2$; $w = z^3$

ii) $x \geq 1$; $w = 1/z$

iii) $-\pi/2 < y < \pi/2$; $w = e^z$

iv) $x \geq 0, y \geq 0, |z| \leq 4$; $w = z^2$

V) the line $x = \frac{\pi}{2}$; $w = \sin z$.

Q.16 Find the LFT that maps $z_1 = -2, z_2 = 0, z_3 = 2$ onto $w_1 = \infty$, $w_2 = 1/4$, $w_3 = 3/8$ respectively. What is the image of x-axis? Also find image of y-axis.

Q.17 Find the image of the circle $x^2 + y^2 = 1$ under $w = \dfrac{5-4z}{4z-2}$

Q.18 Is the mapping $w = \bar{z}$ conformal?

All the best